

CYBERSECURITY CASE STUDY

Site-to-Site IPsec VPN Implementation and Validation

OPNsense • IKEv2 • ESP • Wireshark

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Network Security | VPN Engineering | Packet Analysis

Executive summary

This case study evaluates the confidentiality benefit of a site-to-site IPsec VPN across an untrusted transit network. A baseline routed connection exposed an HTTP request and test-file contents to packet inspection. After an IKEv2 tunnel was deployed between two OPNsense firewalls, the application remained reachable while the transit observer could see only VPN control traffic and encrypted ESP packets.

Outcome

The before-and-after packet captures provided direct evidence: plaintext application data was readable before the VPN, while the same internal conversation was encapsulated and unreadable after IPsec activation.

Environment	Two OPNsense sites, Ubuntu web server, Windows 10 client and Kali transit host.
Core tools	OPNsense, IKEv2/IPsec, Wireshark, curl and virtual networks.
Cryptography	AES-256 encryption, SHA-256 integrity, DH Group 14 and ESP child-tunnel protection.
Security focus	Confidentiality, integrity, segmentation, least-exposure firewalling and validation.
Safety boundary	Authorized isolated lab; the "confidential" file contained synthetic demonstration data.

Network design

The topology represented a headquarters LAN and branch LAN connected through two simulated WAN segments. Kali routed the baseline path and provided the untrusted observation point. OPNsense firewalls became the VPN peers, with the Windows branch client accessing a test resource hosted by Ubuntu at headquarters.

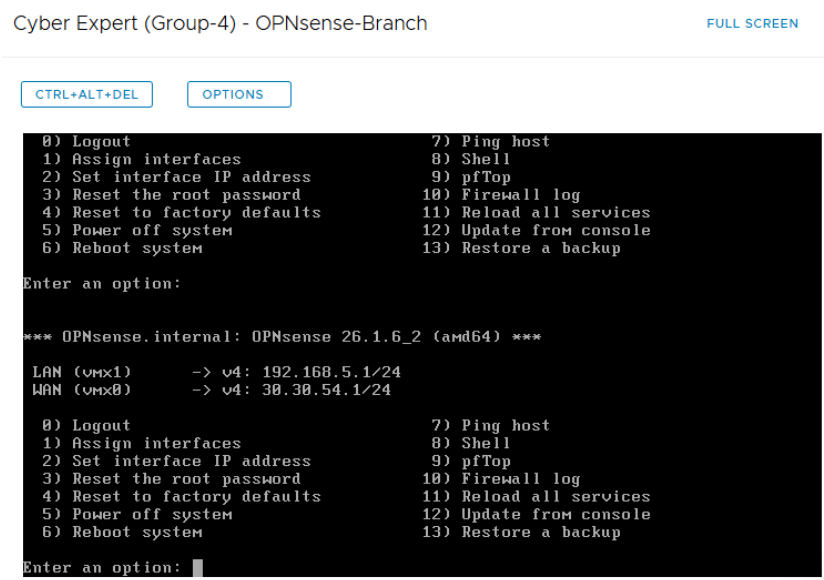


Figure 1. Completed multi-network virtual lab topology.

Baseline: exposed communication

Static routes and narrow test firewall rules first established end-to-end ICMP and HTTP connectivity without encryption. A fresh HTTP request was generated with curl to avoid browser caching and produce a complete response for inspection.

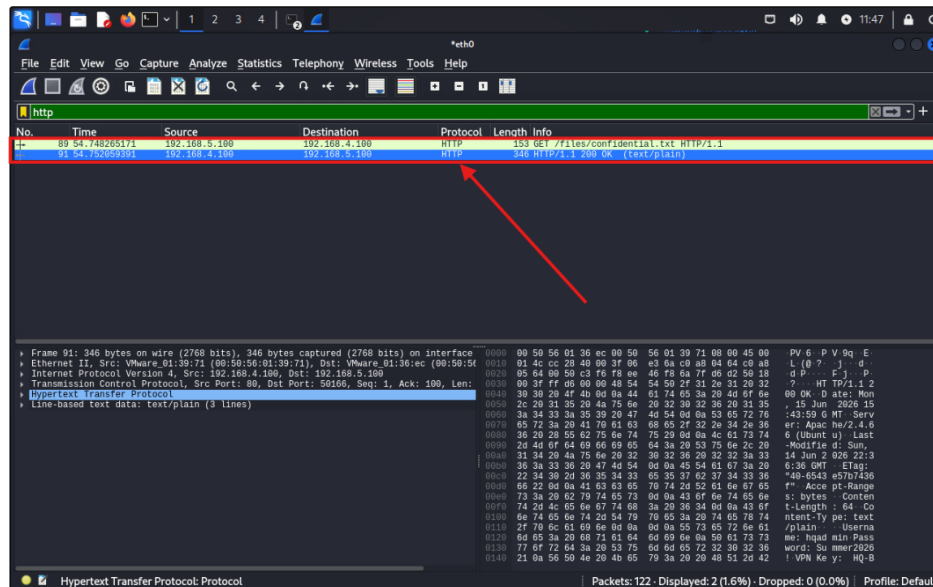


Figure 2. Wireshark inspection of the unencrypted application conversation.

Security finding

The transit observer could identify the internal source and destination, follow the HTTP exchange and read the synthetic file contents. This demonstrated that routing and firewall reachability alone did not provide payload confidentiality.

VPN implementation

- Created IKEv2 connection definitions on the headquarters and branch OPNsense peers.
- Configured mutual pre-shared-key authentication for the controlled lab.
- Applied AES-256, SHA-256 and DH Group 14 proposals, with ESP protecting the internal subnets.
- Restricted WAN exposure to required VPN traffic, including UDP 500, UDP 4500 and ESP from the known peer.
- Added IPsec and LAN rules for the intended branch-to-headquarters application path.

Validation: protected communication

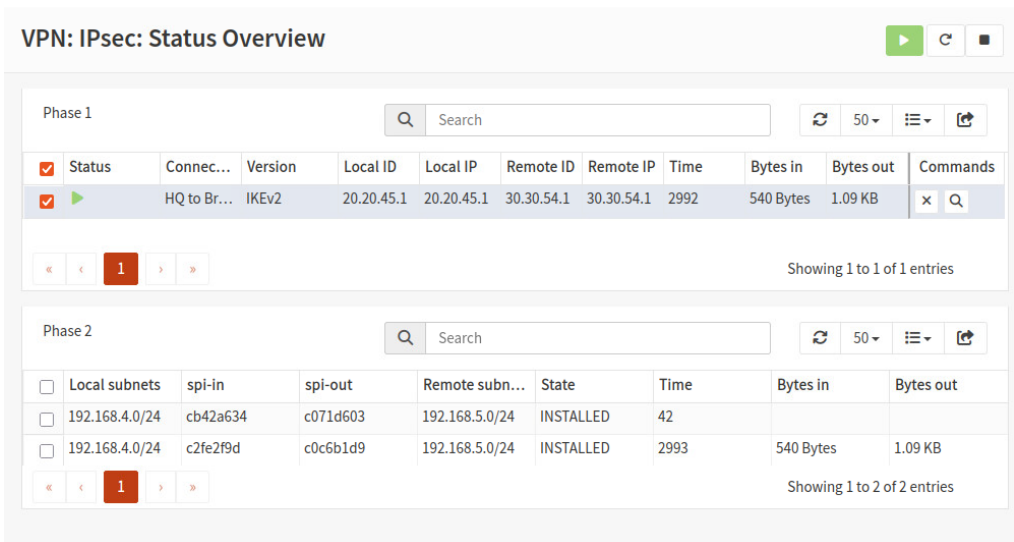


Figure 3. OPNsense IPsec status showing established security associations.

The Windows client retained connectivity to the Ubuntu resource after the tunnel was established. At the transit point, packet inspection showed VPN peer addresses and ESP traffic rather than the internal HTTP payload.

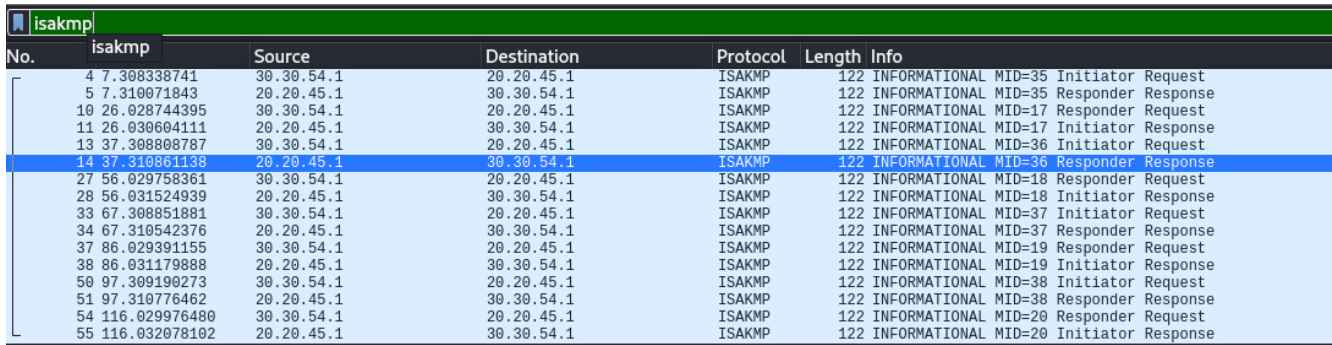


Figure 4. Post-VPN capture showing encrypted ESP traffic across the untrusted segment.

Before-and-after evidence

Before IPsec	Internal host addresses, HTTP request path and test-file content were observable.
After IPsec	Observer saw VPN peer metadata and ESP/IKE traffic; internal application content was not readable.
Availability	The branch client continued reaching the headquarters service through the encrypted path.
Control result	IPsec reduced disclosure risk on the transit network while preserving authorized connectivity.

Troubleshooting and operational judgment

During validation, the tunnel security associations were active but client traffic initially failed. Packet capture and firewall live view isolated the problem to a disabled branch LAN allow rule. Re-enabling the intended rule restored communication while the earlier plaintext path remained disabled. This distinguished tunnel health from policy-path correctness.

Production recommendations

- Replace shared secrets with certificate-based authentication when scale, revocation and device identity matter.
- Use unique credentials per peer and protect key material through controlled lifecycle management.
- Limit VPN-facing firewall rules to known peers and required protocols.
- Monitor tunnel state, authentication failures, configuration changes, latency and packet loss.
- Retire temporary plaintext rules after migration and verify with independent packet capture.
- Plan for metadata exposure and availability threats: IPsec protects payloads but does not hide peer addresses, timing or volume.

Skills demonstrated

Network security	Segmented addressing, routing, firewall policy and inter-site connectivity.
VPN engineering	IKEv2/IPsec peer configuration, cryptographic proposals and traffic selectors.
Packet analysis	Before/after capture, protocol filtering and payload-verification methodology.
Troubleshooting	Layered isolation using tunnel status, packet capture and firewall decision logs.
Risk analysis	Authentication scaling, residual metadata exposure and availability considerations.

Conclusion

The project demonstrated the protective value of a site-to-site VPN with directly observable evidence. The controlled comparison showed that IPsec changed the transit view from readable application traffic to encrypted ESP while preserving the required business connection.

Selected references

OPNsense IPsec Documentation — <https://docs.opnsense.org/manual/how-tos/ipsec-s2s.html>
NIST SP 800-77 Rev. 1, Guide to IPsec VPNs — <https://csrc.nist.gov/pubs/sp/800/77/r1/final>
Wireshark ESP Display Filter — <https://www.wireshark.org/docs/dfref/e/esp.html>